

ISILKUL, Russian Federation

WMO: 286880

Lat: 54.9374N

Lon: 71.2702E

Elev: 415

StdP: 14.48

Time Zone: 6.00 (E06)

Period: 94-18

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -29.1	(c) -24.0	(d) -34.5	(e) 0.8	(f) -28.7	(g) -29.3	(h) 1.1	(i) -23.8	(j) 18.2	(k) 14.1	(l) 16.2	(m) 16.3	(n) 2.6	(o) 250	(p) 0.626

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 18.5	(c) 87.0	(d) 65.9	(e) 83.5	(f) 65.0	(g) 80.4	(h) 63.8	(i) 69.4	(j) 80.7	(k) 67.8	(l) 78.5	(m) 66.2	(n) 76.6	(o) 7.8	(p) 230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.7	96.2	73.8	64.0	90.5	72.5	62.2	85.0	71.2	33.8	80.6	32.4	78.2	31.1	76.7	86.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.8	16.6	14.4	-33.5	92.1	6.8	4.0	-38.4	95.0	-42.4	97.3	-46.2	99.5	-51.2	102.4
			-33.7	72.5	6.8	3.7	-38.5	75.2	-42.5	77.4	-46.3	79.5	-51.2	82.2
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	35.7	0.2	4.3	18.9	39.6	54.6	64.1	65.5	62.2	51.2	38.5	19.7	7.2		
	DBStd	25.83	14.14	12.99	10.96	10.90	9.44	7.92	6.65	7.30	8.22	8.80	14.25	14.36		
	HDD50	6884	1544	1279	964	341	62	5	0	3	81	368	910	1327		
	HDD65	10968	2009	1699	1429	763	342	110	78	142	421	823	1360	1792		
	CDD50	1648	0	0	0	28	203	427	481	382	117	10	0	0		
	CDD65	258	0	0	0	0	18	83	94	56	7	0	0	0		
	CDH74	2727	0	0	0	11	290	877	914	541	94	0	0	0		
	CDH80	783	0	0	0	1	73	286	253	151	19	0	0	0		
(14) Wind	WSAvg	6.9	6.1	6.3	7.3	8.0	8.2	6.8	6.3	6.3	6.5	7.0	7.4	6.9		
(15) Precipitation	PrecAvg	14.5	0.6	0.5	0.7	0.9	1.1	1.9	2.6	2.0	1.1	1.1	1.1	0.8		
	PrecMax	22.1	1.5	1.1	2.0	2.4	2.4	3.8	4.3	4.7	2.3	2.9	2.6	1.7		
	PrecMin	9.6	0.1	0.0	0.0	0.2	0.1	0.6	0.4	0.7	0.3	0.2	0.2	0.2		
	PrecStd	3.1	0.4	0.2	0.5	0.6	0.6	0.7	1.0	1.1	0.6	0.6	0.6	0.4		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	32.5	34.1	45.3	74.6	87.1	90.1	91.4	89.4	82.4	65.4	45.5	34.9		
		MCWB	31.4	32.4	39.4	54.9	62.3	65.6	68.1	66.1	59.4	51.5	41.8	32.4		
	2%	DB	28.9	30.8	38.4	67.3	81.3	86.9	85.8	83.9	75.7	61.0	41.4	32.5		
		MCWB	27.5	29.2	34.9	51.0	59.1	65.6	67.2	65.5	56.9	50.0	39.0	31.1		
	5%	DB	25.5	27.7	35.9	62.0	76.8	83.1	82.3	79.8	69.9	55.9	38.7	29.6		
		MCWB	24.2	26.3	33.3	49.6	57.5	65.0	66.6	64.2	55.8	47.2	36.3	28.1		
	10%	DB	21.5	24.0	33.7	56.6	72.1	79.0	79.1	76.0	65.1	51.8	35.2	25.9		
		MCWB	20.4	22.7	31.5	47.0	55.6	63.2	65.8	63.1	53.9	45.2	33.3	24.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	31.5	32.6	39.7	56.7	64.3	71.0	71.5	70.6	62.1	54.3	42.8	33.0		
		MCDB	32.5	34.0	44.1	71.6	80.7	84.0	82.2	81.4	77.0	61.9	44.7	33.9		
	2%	WB	27.6	29.2	35.5	52.9	61.0	68.6	69.8	68.0	59.3	51.2	39.4	31.0		
		MCDB	28.9	30.6	37.7	64.4	76.4	80.7	80.6	78.3	71.0	58.3	41.4	32.5		
	5%	WB	24.1	26.4	33.6	50.0	58.8	66.4	68.4	66.2	57.2	48.3	36.3	28.3		
		MCDB	25.3	27.7	35.7	60.6	74.1	79.1	78.9	76.2	67.9	55.0	38.3	29.5		
	10%	WB	20.4	22.7	31.6	47.4	56.8	64.4	66.9	64.4	54.8	45.5	33.7	24.4		
		MCDB	21.5	24.1	33.7	56.3	70.0	76.0	76.8	73.4	63.6	51.2	35.1	25.7		
(35) Mean Daily Temperature Range	5% DB	MDBR	13.3	14.8	15.4	16.7	21.4	19.6	18.5	18.9	18.0	13.6	10.6	12.6		
		MCDBR	14.0	12.5	13.6	26.1	28.0	26.0	24.1	25.8	25.6	21.9	10.6	12.6		
	5% WB	MCWBR	13.8	11.8	11.3	13.8	12.1	10.2	10.3	10.8	11.6	13.2	9.3	11.9		
		MCWBR	14.1	12.1	12.5	24.1	25.2	21.9	20.4	20.7	23.0	19.8	10.1	12.3		
(40) Clear-Sky Solar Irradiance	taub		0.284	0.316	0.373	0.411	0.404	0.407	0.432	0.432	0.373	0.354	0.320	0.287		
	taud		2.050	2.019	1.943	2.112	2.228	2.298	2.218	2.185	2.313	2.238	2.104	2.011		
	Ebn at Noon		194	232	244	255	266	266	256	247	248	219	177	159		
	Edh at Noon		22	32	44	43	41	39	41	40	31	26	21	18		
(44) All-Sky Solar Radiation	RadAvg		311	655	1144	1476	1788	1904	1768	1429	997	510	252	201		
	RadStd		23	29	94	129	134	119	128	118	80	70	35	25		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (9 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page